Chapter 4. Inference for Two Populations

Rank-sum test, shift estimation, paired comparisons, scale tests, Kolmogorov–Smirnov

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Overview

Comparing two populations is one of the most basic and most frequent problems in statistics. This chapter uses two running examples.

Example 4.1

Acute leukemia splits into two types, acute myelogenous leukemia (AML) and acute lymphocytic leukemia (ALL). The expression level of a particular gene was measured for 72 acute leukemia patients: $m = 47$ have ALL and $n = 25$ have AML (Golub et al., 1999). The first three rows of the data are shown below; the distributions of the expression level (`ge`) by type are in Figure 4.1 of the book.

```
> leukemia4= read.csv(file = "data/leukemia4.csv")
> head(leukemia4,3)
   ge label
1 328   ALL
2 295   ALL
3 276   ALL
```

Is there a difference in gene expression between the two types ALL and AML?

Example 4.2

To test the hypothesis that the higher the residual food intake (RFI) of Korean cattle, the better the meat quality, the RFI and marbling score (Marb) of 7 cattle were recorded. RFI was classified as high or low.

RFI	Marb
High	740, 610, 720
Low	580, 680, 850, 510

4.1 Differences between two populations

For comparing two populations we assume data

$$X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} F, \qquad Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} G.$$

Of the two models where permutation tests apply (randomization and population models) this is the population model; the technical steps of the test are the same for data from a randomization model, so for simplicity we assume the population model. We are interested in inference about the difference between the unknown populations, which can be described on several levels:

1. **Same shape and dispersion, different location.** For a location shift parameter Δ, $F(x) = G(x + \Delta)$ or $f(x) = g(x + \Delta)$.[1] If the shape $f(x)$ is a well-known distribution (e.g. normal), inference is parametric; with no restriction on the shape it is nonparametric.
2. **Same shape, different dispersion and location.** With a shift Δ and scale $\gamma > 0$, relative to the density $f(x)$ of the first population the second has density $g(x) = \frac{1}{\gamma} f\left(\frac{x-\Delta}{\gamma}\right)$. If $\gamma > 1$, G has the larger variance.
3. **Same shape and location, different dispersion.** The model is $g(x) = \frac{1}{\gamma} f\left(\frac{x}{\gamma}\right)$ for a scale $\gamma > 0$.

[1] If $\Delta > 0$, G lies to the right of F; with the medians θ_F or means μ_F as reference, $\theta_F + \Delta = \theta_G$ and $\mu_F + \Delta = \mu_G$.

4. **Different shapes, possibly with different location or dispersion.** The difference cannot be described by finitely many parameters, but generally the shift or scale parameter is taken as the parameter of interest and a distribution-free test is used. Even under a null hypothesis of equal location or dispersion the two populations may differ.
5. **Different shapes and no parameter of interest.** The hypotheses are not about a parameter: $H_0 : F = G$, $H_1 : F \neq G$.

With $H_0 : F = G$ the exchangeability condition of the permutation test is satisfied. In the location-shift model (case 1) and the scale model (case 3), the null hypotheses $H_0 : \Delta = 0$ and $H_0 : \gamma = 1$ make the two populations equal (case 5 too). In cases 2 and 4, however, $H_0 : \Delta = 0$ does not imply equal populations, and the permutation test cannot be applied.

4.2 Tests for location

4.2.1 The two-sample t-test and permutation tests

First the traditional case where normality can be assumed. The first population F is $N(\mu_F, \sigma_F^2)$ and the second G is $N(\mu_G, \sigma_G^2)$. In this parametric model the location parameter Δ is the difference of means $\Delta = \mu_G - \mu_F$ and the scale parameter γ the ratio of standard deviations $\gamma = \sigma_G/\sigma_F$.

In case 1 above we compare the means of two normal distributions with equal variances, and the two-sample t-test using

$$T = \Big(\frac{1}{n} + \frac{1}{m}\Big)^{-\frac{1}{2}} \frac{\bar{Y}_n - \bar{X}_m}{s_p}$$

is exact and the most powerful. Under H_0, T follows t_{m+n-2}, which gives the p-value.

In case 2 we compare the means of two normal distributions with different variances, and the two-sample (unequal variance) t-test with Welch's statistic

$$T = \frac{\bar{Y}_n - \bar{X}_n}{\sqrt{\frac{s_X^2}{m} + \frac{s_Y^2}{n}}}$$

is known to be approximately exact. Under H_0, T is approximately t_ν, with degrees of freedom approximated by

$$\nu \approx \Big(\frac{s_X^2}{m} + \frac{s_Y^2}{n}\Big)^2 \Big/ \Big(\frac{s_X^4}{m^2(m-1)} + \frac{s_Y^4}{n^2(n-1)}\Big).$$

Permutation tests for location. When normality is badly violated, or the sample is too small for the exactness of the two-sample t-test, a permutation test can be performed, with null hypothesis $H_0 : F = G$. As statistic one can use the sum $\sum_{i=1}^n Y_i$ of the second sample, the two-sample t-statistic T, the difference of medians, or the difference of trimmed means. The permutation test is exact whichever statistic is used, but depending on the shape of the population some statistics are preferred (more powerful); e.g. for asymmetric distributions the difference of medians is more powerful (see Chapter 3).

As in one-sample tests, instead of the observations themselves one can use their ranks, signs or scores and run a permutation test. Traditional nonparametric (distribution-free) methods for two-sample problems were developed on exactly such transformations.

4.2.2 The Wilcoxon rank-sum test

The **rank-sum test** (Wilcoxon, 1945) proposed by Wilcoxon is the most widely used nonparametric method for comparing the locations of two populations. It is often used when the two-sample t-test

is not exact or lacks power, e.g. for heavy-tailed populations. It is a permutation test using the sum of ranks as statistic, instead of comparing means or medians of the two samples.

Wilcoxon's rank-sum statistic is defined as follows. Arrange $X_1, \ldots, X_m$ and $Y_1, \ldots, Y_n$ in order and assign ranks from the smallest value. The statistic is the sum of the ranks of one of the two samples (e.g. the Y's). With S_i $(i = 1, \ldots, m)$ the ranks of the X observations and R_i $(i = 1, \ldots, n)$ those of the Y observations,

$$W = \sum_{i=1}^{n} R_i.$$

Example 4.3

The ranks of the data of Example 4.2:

```
> cowdat = read.csv("data/cow.csv")
> ranks = rank(cowdat$Marb)
> data.frame(cowdat, ranks)
   RFI Marb ranks
1 High  740     6
2 High  610     3
3 High  720     5
4  Low  580     2
5  Low  680     4
6  Low  850     7
7  Low  510     1
```

With ties among the observations, mid-ranks are used (e.g. if there were two smallest values 510, both would get rank 1.5). Taking the `RFI==Low` sample as Y, the observed rank-sum statistic is $W = 2 + 4 + 7 + 1 = 14$:

```
> (W.obs = sum( ranks[cowdat$RFI == "Low"] ))
[1] 14
```

If $H_0 : F = G$ is true, the observations $(X_1, \ldots, X_m, Y_1, \ldots, Y_n)$ are exchangeable, and so are the ranks $(S_1, \ldots, S_m, R_1, \ldots, R_n)$. The null distribution of Wilcoxon's rank-sum statistic W is its permutation distribution over all $\binom{m+n}{n}$ shuffles. For the marbling data of Example 4.2 there are $\binom{7}{4} = 35$ shuffles, and the null distribution of W over all of them is:

```
> allperms.id = gtools::combinations(7,4)
> W.perm = apply(allperms.id,1,
+                function(perm.y.id){ sum(ranks[perm.y.id]) })
> table(W.perm)
```

w	10	11	12	13	14	15	16	17	18	19	20	21	22
$P(W = w)$	$\frac{1}{35}$	$\frac{1}{35}$	$\frac{2}{35}$	$\frac{3}{35}$	$\frac{4}{35}$	$\frac{4}{35}$	$\frac{5}{35}$	$\frac{4}{35}$	$\frac{4}{35}$	$\frac{3}{35}$	$\frac{2}{35}$	$\frac{1}{35}$	$\frac{1}{35}$

(Figure 4.2 of the book plots it.) The value of W depends only on the ranks of the observations, not on their values, so the null distribution of W does not depend on the population. The distribution-free permutation test using this null distribution is Wilcoxon's rank-sum test.

Example 4.4

In Example 4.2 let $H_0 : F = G$ and $H_1 : F(x) \leq G(x)$ (the population with `RFI == High`, "X", lies to the right of the population with `RFI == Low`, "Y"). The smaller the rank sum W of the `RFI == Low` sample, the more extreme under H_0. So the p-value of $W_{\text{obs}} = 14$ is

$P(W \le W_{\text{obs}}) = P(W \le 14) = 11/35$: we cannot be sure that higher RFI improves meat quality.

For reference, using the rank sum $W' = \sum_{i=1}^{m} S_i$ of the other sample (the "X" observations) as statistic gives the same permutation test, since $W + W' = N(N+1)/2$ is constant.

The Mann–Whitney test. With observations $X_1, \ldots, X_m$ from the first population and $Y_1, \ldots, Y_n$ from the second, Mann and Whitney proposed as statistic for the location difference the number of pairs (X_i, Y_j) with Y_j larger than X_i:

$$U = \sum_{i=1}^{m} \sum_{j=1}^{n} I(Y_j > X_i),$$

the **Mann–Whitney U-statistic**. With ties in the data, $I(Y_j > X_i) + \frac{1}{2} I(Y_j = X_i)$ is used instead of $I(Y_j > X_i)$. The U-statistic is closely related to Wilcoxon's rank-sum statistic:

$$W = U + \frac{n(n+1)}{2}.$$

Proof (supplied). Assume no ties. The rank of Y_j in the pooled sample is one plus the number of observations smaller than Y_j:

$$R_j = 1 + \#\{k : Y_k < Y_j\} + \#\{i : X_i < Y_j\}.$$

Summing over j, $\sum_j \#\{k : Y_k < Y_j\} = 0 + 1 + \cdots + (n-1)$ (the jth smallest Y has $j-1$ smaller Y's), so $W = n + \frac{n(n-1)}{2} + U = U + \frac{n(n+1)}{2}$. ■

So the permutation test with U (Mann–Whitney test) gives always the same result as the permutation test with W (Wilcoxon rank-sum test); both are called the **Mann–Whitney–Wilcoxon test**.

The distribution of the Mann–Whitney U is that of W shifted left by $n(n+1)/2$; knowing one gives the other. When $H_0 : F = G$ is true and there are no ties, the distribution of W is called the **Wilcoxon rank-sum distribution**.[2] It is completely determined by m, n and does not depend on the data values, so the Mann–Whitney–Wilcoxon test is distribution-free.

When (m, n) is not large, the exact rank-sum distribution over all $\binom{m+n}{m}$ shuffles is easily computed; for large samples the number of shuffles is huge and exact p-values take very long. Then one may use a random permutation test, but the usual approach is the large-sample approximation of the rank-sum distribution introduced next.

4.2.3 Large-sample approximation

In the leukemia gene-expression data of Example 4.1, let the ALL expression levels be $X_1, \ldots, X_m \sim F$ and the AML ones $Y_1, \ldots, Y_n \sim G$, and apply the rank-sum test. Here $m = 47$, $n = 25$ are too large to consider all shuffles for the null distribution. The rank sum $W = \sum_{i=1}^{n} R_i$ is of the form of a "mean" of ranks, so the CLT can approximate its distribution by a normal for large m, n.

Recall the assumptions of the CLT. The most basic CLT says that for i.i.d. observations $Z_1, \ldots, Z_n$ the sample mean is approximately normal for large n (Theorem 1.1):

$$\bar{Z}_n = \frac{1}{n} \sum_{i=1}^{n} Z_i \dot{\sim} N\Big(\mu_Z, \frac{\sigma_Z^2}{n}\Big), \tag{4.1}$$

[2] The distribution of W is sometimes called the Wilcoxon rank-sum distribution and so is that of U; one determines the other. Here (no ties) we call the distribution of W the Wilcoxon rank-sum distribution. R's `dwilcox()`, `qwilcox()` etc. give the distribution of U for small m, n.

with $E(Z_1) = \mu_z$ and $\text{Var}(Z_1) = \sigma_Z^2$. Let $N = m + n$. In the rank-sum statistic, under H_0 the ranks $R_1, \ldots, R_n$ all have the same distribution (discrete uniform on $\{1, \ldots, N\}$). But if $R_1 = 1$, R_2 cannot be 1, so the ranks are not independent and the basic CLT (4.1) does not apply. Instead, the ranks $R_1, \ldots, R_n$ can be seen as a sample drawn *without replacement* with equal probabilities from the finite population $\{1, \ldots, N\}$. The rank sum W is the sum (mean $\times n$) of a sample of size n drawn without replacement from a finite population, and a CLT for finite populations applies.

For reference, the basic CLT (4.1) applies to means of samples drawn *with replacement*, whether the population is infinite or finite.

Samples from a finite population. Consider a finite population $\mathcal{A}_N = \{A_1, \ldots, A_N\}$ of N units. Each A_i is a number and may take any fixed value; e.g. ranks of observations $A_1 = 1, \ldots, A_N = N$, or with ties several units with the same value, $A_1 = 1.5$, $A_2 = 1.5$, $A_3 = 3, \ldots, A_N = N$. The population mean and variance of this finite population are

$$\mu = \frac{1}{N}\sum_{i=1}^{N} A_i \equiv \bar{A}, \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(A_i - \mu)^2 = \frac{\sum_{i=1}^{N} A_i^2}{N} - (\bar{A})^2 \equiv \overline{A^2} - (\bar{A})^2. \tag{4.2}$$

If X is drawn at random from the discrete uniform distribution on $\mathcal{A}_N$, then $\mu = E(X)$ and $\text{Var}(X) = \sigma^2$.

Sampling with replacement. If $X_1, \ldots, X_n$ are drawn with replacement from $\mathcal{A}_N$,

$$X_i \overset{\text{i.i.d.}}{\sim} \text{Uniform}(\mathcal{A}_N), \tag{4.3}$$

so $E(\bar{X}_n) = \mu$, $\text{Var}(\bar{X}_n) = \sigma^2/n$, and by the basic CLT (4.1), for large n, $\bar{X}_n \dot{\sim} N(\mu, \sigma^2/n)$, i.e. as $n \to \infty$

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$

Sampling without replacement. If $X_1, \ldots, X_n$ are drawn without replacement from $\mathcal{A}_N$, the sample $\{X_1, \ldots, X_n\}$ is one of the $\binom{N}{n}$ ways of choosing n of the N units $A_1, \ldots, A_N$. The distribution of $\{X_1, \ldots, X_n\}$ is uniform over all size-n subsets of $\mathcal{A}_N$: for every subset $\{p_1, \ldots, p_n\} \subset \{1, \ldots, N\}$, the probability that the sample is $\{A_{p_1}, \ldots, A_{p_n}\}$ is

$$P(\{X_1, \ldots, X_n\} = \{A_{p_1}, A_{p_2}, \ldots, A_{p_n}\}) = \frac{1}{\binom{N}{n}}.$$

The sample mean $\bar{X}_n$ of a sample drawn without replacement is a good estimator of the population mean μ.

Theorem 4.1

Let $X_1, \ldots, X_n$ be drawn without replacement from a finite population $\mathcal{A}_N$ of N units with mean μ and variance σ^2, and let $\bar{X}_n$ be the sample mean. Then

$$E(\bar{X}_n) = \mu, \qquad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}\frac{N-n}{N-1}.$$

Proof. The values of $X_1, \ldots, X_n$ are n of $A_1, \ldots, A_N$. Let $Z_i = 1$ if A_i is among the sample values and $Z_i = 0$ otherwise. Then $X_1 + \cdots + X_n \overset{d}{=} \sum_{i=1}^{N} Z_i A_i$. The indicators $(Z_1, \ldots, Z_N)$ are not independent, but each Z_i has the same marginal distribution with $P(Z_i = 1) = n/N$. So

$$E(\bar{X}_n) = \frac{1}{n}\sum_{i=1}^{N} A_i P(Z_i = 1) = \frac{1}{n} \cdot \frac{n}{N}\sum_i A_i = \mu.$$

For the variance we need the joint distribution of (Z_i, Z_j). (Supplied.) For $i \neq j$, $P(Z_i = Z_j = 1) = \binom{N-2}{n-2}/\binom{N}{n} = \frac{n(n-1)}{N(N-1)}$, so

$$\operatorname{Var}(Z_i) = \frac{n}{N}\Big(1 - \frac{n}{N}\Big), \qquad \operatorname{Cov}(Z_i, Z_j) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n}{N} \cdot \frac{N-n}{N(N-1)}.$$

The variance of $\bar{X}_n$ is unchanged by shifting all A_i by a constant, so take $\mu = 0$; then $\sum_i A_i^2 = N\sigma^2$ and $\sum_{i \neq j} A_i A_j = (\sum_i A_i)^2 - \sum_i A_i^2 = -N\sigma^2$. Hence

$$\operatorname{Var}\Big(\sum_i Z_i A_i\Big) = N\sigma^2 \cdot \frac{n}{N}\frac{N-n}{N} + N\sigma^2 \cdot \frac{n}{N}\frac{N-n}{N(N-1)} = n\sigma^2 \frac{N-n}{N}\Big(1 + \frac{1}{N-1}\Big) = n\sigma^2 \frac{N-n}{N-1},$$

and dividing by n^2 gives the claim. ■

Compared with the variance σ^2/n of the mean of a sample drawn with replacement, the variance of the mean of a sample drawn without replacement is smaller by the factor $\frac{N-n}{N-1}$, the **finite-population correction factor**. For sample size $n = 1$, $\frac{N-n}{N-1} = 1$, and it decreases as n grows. For $n = N$ (the whole population drawn) the variance of the sample mean must be 0, and indeed $\frac{N-n}{N-1} = 0$.

The sample size n is always smaller than N, so for a large-sample approximation of $\bar{X}_n$, N must grow along with n. The CLT for sample means drawn without replacement from a finite population:

Theorem 4.2 (Hájek (1960); Li and Ding (2017))

For every $N = 1, 2, 3, \ldots$ let a finite population $\mathcal{A}_N = \{A_{N1}, \ldots, A_{NN}\}$ be given, with mean μ_N and variance σ_N^2. Let $\bar{X}_n$ be the mean of a sample $X_1, \ldots, X_n$ drawn without replacement from $\mathcal{A}_N$. If $\min\{n, N-n\} \to \infty$ and (Hájek's condition, supplied)

$$\frac{\max_i (A_{Ni} - \mu_N)^2}{\min\{n, N-n\}\,\sigma_N^2} \to 0,$$

then

$$\frac{\bar{X}_n - E(\bar{X}_n)}{\sqrt{\operatorname{Var}(\bar{X}_n)}} = \frac{\bar{X}_n - \mu_N}{\sigma_N/\sqrt{n}}\sqrt{\frac{N-1}{N-n}} \xrightarrow{d} N(0,1).$$

Proof (sketch, supplied). Without the displayed condition the statement fails (e.g. a population with one huge value), so some such condition is needed; for ranks $A_{Ni} = i$ it holds, since $\max_i (i - \frac{N+1}{2})^2 \le N^2/4$ while $\sigma_N^2 = (N^2-1)/12$. Write $S = \sum_i Z_i (A_{Ni} - \mu_N)$ with the inclusion indicators Z_i of Theorem 4.1. Hájek couples sampling without replacement with *Poisson sampling*: let $I_1, \ldots, I_N$ be i.i.d. Bernoulli(n/N) and $S^* = \sum_i I_i (A_{Ni} - \mu_N)$. S^* is a sum of independent terms with variance $\frac{n}{N}(1 - \frac{n}{N})N\sigma_N^2$, and the displayed condition is exactly Lindeberg's condition for it, so $S^*/\operatorname{sd}(S^*) \xrightarrow{d} N(0,1)$. Given $K = \sum_i I_i$, the selected set under Poisson sampling is a uniform K-subset. One adds or removes $|K - n|$ randomly chosen units to obtain a uniform n-subset, i.e. a realization of S. Since $|K - n| = O_p(\sqrt{n(N-n)/N})$, one shows $E(S - S^*)^2/\operatorname{Var}(S) \to 0$. Hence $S/\operatorname{sd}(S)$ has the same normal limit, and $\operatorname{Var}(S) = n^2 \operatorname{Var}(\bar{X}_n)$ by Theorem 4.1. ■

Large-sample approximation of the Wilcoxon rank-sum distribution. The rank sum $W = \sum_{i=1}^n R_i$ is the sum of n observations drawn without replacement from the finite population $\mathcal{A}_N = \{1, \ldots, N\}$. Under H_0 this is simple random sampling without replacement, and by (4.2) the population mean and variance of $\mathcal{A}_N$ are

$$\mu_N = \frac{N+1}{2}, \qquad \sigma_N^2 = \frac{(N+1)(2N+1)}{6} - \Big(\frac{N+1}{2}\Big)^2 = \frac{N^2-1}{12}.$$

The rank sum is related to the mean rank by $W = n\bar{R}$, so by Theorem 4.1

$$E(W) = n\mu = \frac{n(N+1)}{2}, \qquad \mathrm{Var}(W) = n^2 \frac{\sigma^2}{n} \frac{N-n}{N-1} = \frac{nm(N+1)}{12},$$

and by Theorem 4.2, for large n and m,

$$\frac{W - E(W)}{\sqrt{\mathrm{Var}(W)}} \dot{\sim} N(0,1). \tag{4.4}$$

So the null distribution of W can be approximated by the normal $N(E(W), \mathrm{Var}(W))$.

With ties among the observations the ranks considered are not $\{1, \ldots, N\}$ but, e.g.,

$$\mathcal{A}'_N = \{1.5, 1.5, 4, 4, 4, \ldots, N\}.$$

Then $\mathcal{A}'_N$ is taken as the population, $\mathrm{Var}(W)$ is recomputed, and the CLT is applied. The mean is still $E(W) = n(N+1)/2$ as without ties.

Example 4.5

In the leukemia data of Example 4.1, $m = \#ALL = 47$ and $n = \#AML = 25$, large enough. There are tied observations, and the rank-sum statistic is $W_{\mathrm{obs}} = 1113.5$. The approximate null distribution by (4.4) for the data with ties has $E(W) = 912.5$, $\mathrm{Var}(W) = 7147.917$:

$$W \dot{\sim} N(912.5, 7147.917). \tag{4.5}$$

(Figure 4.3 of the book shows this normal-approximated null distribution with the observed value 1113.5.) The two-sided p-value by the normal approximation is

$$p_G = P\Big(\frac{|W - E(W)|}{\sqrt{\mathrm{Var}(W)}} \geq \frac{|W_{\mathrm{obs}} - E(W)|}{\sqrt{\mathrm{Var}(W)}}\Big) \approx 2P(Z \geq 2.37) = 0.017.$$

For comparison, the random permutation test with the rank-sum statistic: from $M = 1999$ random permutations (approximate) null distribution, 14 of the shuffled statistics w_i were larger than W_{obs}, so

$$P(W \geq W_{\mathrm{obs}}) \approx \frac{1 + \sum_{i=1}^{M} I(w_i \geq W_{\mathrm{obs}})}{M+1} = \frac{15}{2000},$$

and by symmetry of the permutation distribution

$$p_{RP} = P(|W - E(W)| \geq |W_{\mathrm{obs}} - E(W)|) = 2P(W \geq W_{\mathrm{obs}}) \approx \frac{30}{2000} = 0.015,$$

almost the same as the normal approximation.

Large-sample approximation of permutation distributions. In two-sample permutation tests the statistic is often the mean of the observations sampled from one of the populations. The rank-sum statistic is n times the mean of the ranks of the "Y" sample; the statistic $\bar{D} = \frac{D}{n} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ (the mean of the second sample) used to introduce permutation tests is also the mean of n observations drawn without replacement. So the permutation distribution of $\bar{D}$ can be approximated by the CLT. The finite population is the pooled sample $\{X_1, \ldots, X_m, Y_1, \ldots, Y_n\}$, whose population mean is the pooled mean and whose population variance is the variance of the pooled sample without bias correction:

$$\mu = \frac{\sum_i^m X_i + \sum_i^n Y_i}{m+n}, \qquad \sigma^2 = \frac{\sum_{i=1}^m (X_i - \mu)^2 + \sum_{i=1}^n (Y_i - \mu)^2}{m+n}.$$

The permutation distribution of $\bar{D}$ has mean $E(\bar{D}) = \mu$ and variance $\text{Var}(\bar{D}) = \frac{\sigma^2}{n}\frac{m}{m+n-1}$, so for large m, n

$$\frac{\bar{D}-\mu}{\sigma/\sqrt{n}}\sqrt{\frac{N-1}{N-n}} \xrightarrow{d} N(0,1). \tag{4.6}$$

Example 4.6

For the leukemia data of Example 4.1, a random permutation test with $\bar{D} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ (the mean expression score of AML patients) gives the estimated permutation distribution shown as a histogram in Figure 4.4 of the book. Compared with the observed $\bar{D}_{\text{obs}} = 233.12$, 17 of the $B = 1999$ randomly generated $\bar{d}_i$ exceed $\bar{D}_{\text{obs}}$. Assuming the permutation distribution is symmetric, the p-value is

$$p \approx 2 \times (1+17)/2000 = 0.018 :$$

enough evidence that the expression scores of AML and ALL patients differ. The approximate null distribution of $\bar{D}$ from the large-sample approximation (4.6) closely approximates the permutation distribution.

The permutation distribution of $\bar{D}$ is well approximated by a normal because $\bar{D}$ is a mean. Is the permutation distribution of the difference of means $\bar{Y}_n - \bar{X}_m$ or of the t-statistic also well approximated by a normal? If so, what are the mean and variance of that permutation distribution?[3] Even when the exact asymptotic distribution is unknown, if a normal approximation is expected, one can approximate its two parameters μ_n, σ_n^2.

If the permutation distribution is approximately $N(\mu_n, \sigma_n^2)$, the sample mean and variance of the statistics $\{\bar{d}_1, \ldots, \bar{d}_B\}$ from random shuffles estimate μ_n, σ_n^2. The dotted density in Figure 4.4 is the normal approximated from the random permutation distribution,

$$N(\hat{\mu}_D, \hat{\sigma}_D^2), \qquad \hat{\mu}_D = \frac{1}{M}\sum_{i=1}^{M}\bar{d}_i, \quad \hat{\sigma}_D^2 = \frac{1}{M-1}\sum_{i=1}^{M}(\bar{d}_i - \hat{\mu}_D)^2,$$

very similar to the normal from the large-sample approximation.

In the same way the permutation distribution of the t-statistic can be approximated by a normal. Figure 4.5 of the book compares the permutation distribution of the t-statistic generated by random shuffles (histogram) with the approximating distribution (dotted density). If the mean and variance of the permutation distribution are μ_{RP}, σ_{RP}^2, the approximating distribution is $N(\mu_{RP}, \sigma_{RP}^2)$. The observed t-statistic is $T_{\text{obs}} = 2.51$; the random permutation p-value is $p_{RP} = \{1 + \sum_{i=1}^{M} I(|T_i| \geq |T_{\text{obs}}|)\}/(M+1) = 0.0105$, and the approximate p-value is $p_N = P(|Z| \geq |T_{\text{obs}} - \mu_{RP}|/\sigma_{RP}) \approx 0.014$. At the 5% level this gives the same conclusion as Example 4.6.

When the permutation test statistic is a mean, a difference of means, a standardized difference of means, or some function of them, its large-sample distribution is normal, so the permutation distribution can be approximated by a normal. If the large-sample distribution of the statistic is not normal, there is no guarantee. Also, keep in mind that large-sample approximations are for large samples.

4.2.4 Estimating the shift parameter and confidence intervals

In the location-shift model, where the two populations have the same shape and dispersion but different locations, assume $X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} F(x)$ and $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} F(x-\Delta)$ for a shift parameter

[3] With random partitions of a finite population into "Y" and "X" samples, the difference of means $\bar{Y}_n - \bar{X}_m$ is not like sampling with replacement from an infinite population, where $\bar{Y}_n$ and $\bar{X}_m$ are independent. To get $\text{Var}(\bar{Y}_n - \bar{X}_m)$ one must compute $\text{Corr}(\bar{Y}_n, \bar{X}_m)$ (by Lehmann (1975), $\text{Corr}(\bar{Y}_n, \bar{X}_m) = -1$: the total is fixed, so $m\bar{X}_m + n\bar{Y}_n$ is constant). The asymptotic distribution of the t-statistic under sampling from a finite population also requires care.

Δ; there is no restriction on the shape of F. If F has a mean, the mean $E(Y_1)$ of the second population exceeds the mean $E(X_1)$ of the first by Δ. In Section 4.2.2 we saw that the Mann–Whitney–Wilcoxon test of $H_0 : \Delta = 0$ against $H_1 : \Delta < 0$ (or $\Delta > 0$, $\Delta \neq 0$) is distribution-free.

Here we use the Mann–Whitney–Wilcoxon test to estimate Δ (point and interval estimation). Of the two forms of the statistic (Wilcoxon's rank sum and Mann–Whitney's U), we use Mann and Whitney's

$$U = U(0) = \sum_{i=1}^{m} \sum_{j=1}^{n} I(Y_j - X_i > 0).$$

There are mn differences $D_{ij} = Y_j - X_i$; denote them $\{D_1, \ldots, D_{mn}\}$. The Mann–Whitney U is the number of the mn differences D_i that are positive (compared with 0). For a value Δ_0 of the shift,

$$U(\Delta_0) = \sum_{i=1}^{m} \sum_{j=1}^{n} I(Y_j - X_i > \Delta_0) = \sum_{i=1}^{mn} I(D_i > \Delta_0),$$

which tests the more general null hypothesis $H_0 : \Delta = \Delta_0$. As in Section 2.2.4, where the Wilcoxon signed-rank statistic for a single population's location was the sign statistic of the Walsh averages relative to θ_0 and gave a confidence interval, here we use the confidence interval given by the number of differences D_i greater than Δ_0.

With $D_{(k)}$ the kth smallest difference, choose two integers $1 \leq k_l < k_u \leq mn$ and set the interval for Δ as

$$D_{(k_l)} \leq \Delta < D_{(k_u)}, \tag{4.7}$$

choosing k_l, k_u so that $P(D_{(k_l)} \leq \Delta < D_{(k_u)}) \geq 1 - \alpha$ for the level $1 - \alpha$, where Δ is the true unknown shift. Using the relation between $U(\Delta)$ and the D_i, and the facts that $U(\Delta)$ always has the distribution of the Mann–Whitney U whatever the value of Δ and that this distribution is symmetric about $E(U) = mn/2$, one shows

$$P(D_{(k_l)} \leq \Delta < D_{(k_u)}) = P(k_l \leq U(\Delta) \leq k_u - 1) \geq 1 - \alpha. \tag{4.8}$$

Proof of (4.8) (supplied). $D_{(k_l)} \leq \Delta$ iff at least k_l differences are $\leq \Delta$, i.e. $U(\Delta) = \#\{D_i > \Delta\} \leq mn - k_l$; and $\Delta < D_{(k_u)}$ iff at most $k_u - 1$ differences are $\leq \Delta$, i.e. $U(\Delta) \geq mn - k_u + 1$. At the true Δ, $Y_j - \Delta \stackrel{d}{=} X_i$, so $U(\Delta)$ is the Mann–Whitney statistic of the samples X and $Y - \Delta$, which come from the same population: $U(\Delta)$ has the null distribution of U. That distribution is symmetric about $mn/2$, i.e. $mn - U \stackrel{d}{=} U$, so $P(mn - k_u + 1 \leq U(\Delta) \leq mn - k_l) = P(k_l \leq U(\Delta) \leq k_u - 1)$. ■

As for the signed-rank interval for location, with the quantile function Q_U of the null distribution of U, setting

$$k_l = Q_U(\alpha/2), \qquad k_u = mn - k_l + 1 = mn - Q_U(\alpha/2) + 1$$

makes the level of the interval (4.7) at least $1 - \alpha$. As the level of (4.7) is made smaller and smaller, what remains is the median of the differences D_i,

$$\hat{\Delta} := \text{median}\{D_1, \ldots, D_{mn}\} = \text{median}\{Y_j - X_i : i = 1, \ldots, m,\ j = 1, \ldots, n\},$$

the Hodges–Lehmann estimator of Δ.

Example 4.7

In Example 4.2, assume $X_i \stackrel{d}{=} Y_j - \Delta$; find the Hodges–Lehmann estimate of Δ and a two-sided interval at level 80%. First compute and sort the differences $D_{ij} = Y_j - X_i$:

```
> x = cowdat$Marb[cowdat$RFI == "High"]
> y = cowdat$Marb[cowdat$RFI == "Low"]
> m = length(x); n = length(y)
```

```
> D = sapply(y, function(y){y-x}) # Delta = E(Y) - E(X)
> (D.ord = sort(D))
 [1] -230 -210 -160 -140 -100  -60  -40  -30   70  110  130  240
```

All 12 differences: their median, the Hodges–Lehmann estimate, is $\hat{\Delta} = -50$. The interval (4.7) is computed at $\alpha = 0.2$; the quantile of U comes from `qwilcox()`:

```
> alpha = 0.2
> kl = qwilcox(alpha/2,m,n)
> ku = m*n - kl + 1
> c(kl,ku)
[1]  2 11
> (ci = D.ord[c(kl,ku)])
[1] -210  130
```

The 80% interval for Δ is $\Delta = E(Y) - E(X) \in [-210, 130)$. By (4.8) its coverage probability comfortably exceeds 80%:

```
> (coverage = sum(dwilcox(kl:(ku-1),m,n)))
[1] 0.8857143
```

With such small samples no inference can guarantee much accuracy.

Large-sample intervals. For large m, n the quantile function $Q_U(\cdot)$ of U is hard to compute exactly; since U is approximately normal, the normal quantile function approximates $Q_U(\cdot)$. Using $U = W - n(n+1)/2$,

$$\mu_U := E(U) = E(W) - \frac{n(n+1)}{2} = \frac{nm}{2}, \qquad \sigma_U^2 := \mathrm{Var}(U) = \mathrm{Var}(W) = \frac{nm(n+m+1)}{12},$$

so $U \dot{\sim} N(\mu_U, \sigma_U^2)$ and $Q_U(\alpha)$ for $\alpha \in (0,1)$ is approximated by the α-quantile of $N(\mu_U, \sigma_U^2)$:

$$\alpha \le P(U \le Q_U(\alpha)) = P\Big(\frac{U-\mu_U}{\sigma_U} \le \frac{Q_U(\alpha)-\mu_U}{\sigma_U}\Big) \approx P(Z \le \Phi^{-1}(\alpha)).$$

With $\Phi^{-1}(\alpha) = -z_\alpha$,

$$Q_U(\alpha) \approx \mu_U - \sigma_U z_\alpha = \frac{nm}{2} - z_\alpha\sqrt{\frac{nm(n+m+1)}{12}}.$$

With the large-sample orders $\tilde{k}_l = \lfloor \frac{nm}{2} - z_{\alpha/2}\sqrt{\frac{nm(n+m+1)}{12}} \rfloor$ and $\tilde{k}_u = \lfloor \frac{nm}{2} + z_{\alpha/2}\sqrt{\frac{nm(n+m+1)}{12}} + 1 \rfloor$, the interval $D_{(\tilde{k}_l)} \le \Delta < D_{(\tilde{k}_u)}$ is an asymptotically exact $1-\alpha$ interval for the shift Δ.

Example 4.8

In Example 4.1, assume $X_i \overset{d}{=} Y_j - \Delta$ and find a two-sided 95% interval for Δ:

```
> # confidence interval large sample
> x = leukemia4$ge[leukemia4$label == "ALL"]
> y = leukemia4$ge[leukemia4$label == "AML"]
> D = sapply(y, function(y){y-x})
> D.ord = sort(D)
```

There are 1175 differences, not printed. The orders $\tilde{k}_l, \tilde{k}_u$ are generally not integers, so the nearest integers are used:

```
> m = length(x); n = length(y)
> ew = n * (n+m + 1)/2 # mean of W
```

```
> vw = n*m*(n+m+1)/12  # variance of W
> eu = ew - n*(n+1)/2  # mean of U
> vu = vw              # variance of U
> (ci.limits = round( eu + 1.96 * sqrt(vu) *c(-1,1)))
[1] 422 753
> (ci = D.ord[ci.limits])
[1]  13 133
```

The 95% interval for the shift is $\Delta \in (13, 133)$: the gene expression of the AML population is significantly higher than that of the ALL population.

4.2.5 Summary: tests for location

Wilcoxon proposed the rank-sum test (Wilcoxon, 1945) together with the signed-rank test, and Mann and Whitney (1947) studied the properties of the rank-sum distribution and its normal approximation in the form of the U-statistic. Nearly 80 years later, the Mann–Whitney–Wilcoxon test is still the most representative nonparametric test for the locations of two samples. That it is a permutation test using a particular transformation of the data (to ranks) was already used by Wilcoxon, but the term "permutation test" was not used: it had been reserved for tests that shuffle the labels while keeping the observed values (i.e. permutation tests using $\bar{Y} - \bar{X}$ and the like). Today "permutation test" generally refers to methods using as null distribution the permutation distribution conditional on (with the original values fixed) the observed values.

Permutation tests are conditionally exact. In particular, the Mann–Whitney–Wilcoxon test considers only the relative positions of the data, not their values, so it gives the same result whatever population generated the data, as long as the order is the same: a distribution-free method. But the type I error is controlled exactly only if the two populations are completely identical under the null ($H_0 : F = G$). If the two populations have the same median but differ in dispersion or shape, the Wilcoxon test does not guarantee exactness.[4]

Under the null hypothesis, all observations X_i are an independent sample from F, so $U_i = F(X_i) \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0, 1)$. Multiplying the transformed U_i by $N + 1$ and computing the expectations of their order statistics gives $1, 2, \ldots, N$, i.e. the ranks. Using this, the Wilcoxon test can be understood as a permutation test that first transforms the unknown distribution F into the uniform distribution and uses the expectations of the transformed values (ranks) instead of the observations X_i. In this process the actual population distribution F is used indirectly, which is why the null distribution is the same for every population: a distribution-free method.

4.2.6 The R function `wilcox.test()`

The Mann–Whitney–Wilcoxon test of the location difference of two populations is implemented in R's `wilcox.test()`, also used for the one-sample Wilcoxon signed-rank test (see Section 2.2.5). Some usages follow. With the `leukemia4` data frame of Example 4.1, let x_i be the values of `ge` with `label == "ALL"` and y_i those with `label == "AML"`; in the model $x_i = \mu_1 + \epsilon_i$, $y_i = \mu_2 + e_i$ we run the rank-sum test of

$$H_0 : \mu_1 = \mu_2 \quad \text{vs} \quad H_1 : \mu_1 \neq \mu_2.$$

[4] When two populations with the same shape and median but different dispersion are compared, Fligner and Policello (1981) showed that a test using a standardized form of the Wilcoxon rank-sum statistic is approximately exact (the Fligner–Policello test). More recently Chung and Romano (2016) dealt with this situation, showing that a method similar to the derivation of Welch's t-test is approximately exact even under different shapes and medians. Without rank transformation, Yuen's (1974) Yuen–Welch t-test, which uses trimmed means in Welch's t-test, is also known to be approximately exact when normality is badly violated.

The argument `conf.int = TRUE` also computes a 95% confidence interval for the shift $\Delta = \mu_1 - \mu_2$[5]; with `exact = FALSE` the large-sample approximation is used.

```
> wilcox.test(ge ~ label, exact = FALSE, conf.int = TRUE, data =
    leukemia4)

        Wilcoxon rank sum test with continuity correction

data:  ge by label
W = 386.5, p-value = 0.01771
alternative hypothesis: true location shift is not equal to 0
95 percent confidence interval:
 -133.99994  -12.99998
sample estimates:
difference in location
             -68.00001
```

With no ties and small samples, the exact test and interval use the Wilcoxon rank-sum distribution as below. These data have ties, so the test is not exact and a warning appears.

```
wilcox.test(ge ~ label, exact = TRUE, data = leukemia4)
```

It can also be used as follows, with the observations of the two samples in vectors `x` and `y`. Adding `alternative = "greater"` performs the upper-tailed test ($H_1 : \mu_1 < \mu_2$); "greater" is interpreted as $y > x$ in the order of the function's arguments.

```
> wilcox.test(y,x, exact = FALSE, correct = FALSE, alternative = "
    greater")

        Wilcoxon rank sum test

data:  y and x
W = 788.5, p-value = 0.008715
alternative hypothesis: true location shift is greater than 0
```

4.3 Paired two-sample comparisons

Paired two-sample comparisons are best understood as tests about one population.

Example 4.9

A cosmetics company developed a new skin-protection cream. To compare the effect of the new cream (A) with the existing cream (B), 8 users were selected; cream A was applied to one arm and cream B to the other, the skin was exposed to sunlight for a fixed time, and the degree of burn was measured. Can cream A be said to protect the skin better than cream B? (Higher values mean a more severe burn.)

	1	2	3	4	5	6	7	8
A	72	109	74	33	54	62	41	18
B	34	72	165	14	92	75	58	85

In this situation we compare populations A and B, but each observation of A is paired with one of B. User 1 used both creams A and B, so the two observations 72, 34 cannot be regarded as

[5] In Sections 3.6.1, 4.1 and 4.2.4 the shift was defined the other way, $\Delta = \mu_2 - \mu_1$.

independent. Comparing the location (mean or median) of population A with that of population B in this situation is a **paired comparison**.

With observations $X_1, \ldots, X_n, Y_1, \ldots, Y_n$, each (X_i, Y_i) comes from the ith user and so is not independent, but the pairs

$$\{(X_1, Y_1), \ldots, (X_n, Y_n)\}$$

are independent. So assume that the differences $D_i = Y_i - X_i$ of the two effects for each user satisfy

$$D_i = \Delta + \epsilon_i, \qquad i = 1, \ldots, n,$$

with the ϵ_i i.i.d. The distribution of ϵ_i (median 0) may or may not be normal. If each population is normal, the distribution of D_i (and of ϵ_i) is normal too; but if the populations are not normal, or the two distributions differ, the distribution of ϵ_i cannot be taken to be normal.

In a paired comparison we test, for the location difference Δ of the two populations,

$$H_0 : \Delta = 0 \quad \text{vs} \quad H_1 : \Delta \neq 0.$$

(One-sided tests $H_1 : \Delta < 0$ or $H_1 : \Delta > 0$ are possible too.) Since the differences D_i are a sample from one population, a location test for one population is performed: the one-sample t-test under normality, otherwise the sign test or the Wilcoxon signed-rank test. See Chapter 2.

4.4 Tests for the scale parameter

So far we have looked at tests of the locations of two populations and distribution-free estimation of the shift Δ. Here we look at distribution-free tests of differences in dispersion. Inference about dispersion is less common, but when normality cannot be assumed, rank-based distribution-free tests can be considered.

4.4.1 When the two locations are equal

First assume the two populations have the same location and differ only in scale: $X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} f$, $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} g$, with

$$X_i = \mu + \sigma_1 e_i, \qquad Y_i = \mu + \sigma_2 \epsilon_i,$$

where $e_1, \ldots, e_m, \epsilon_1, \ldots, \epsilon_n$ are independent with the same distribution with median 0. The two populations share the location μ and the ratio of dispersions is $\gamma = \sigma_2/\sigma_1$: the dispersion of the second population is γ times that of the first. Figure 4.6 of the book shows the two distributions when the distribution of ϵ and e is normal and $\gamma = 2$.

Example 4.10

A food factory got complaints that the package weights of curry vary from product to product, and wants to introduce a new process to keep the weights constant. Randomly sampled weights from the old process were $110, 115, 120, 105, 103, 117$ g, and from the new process $109, 118, 113, 112, 119$ g. Does the new process keep the weights more constant than the old one?

Assume the mean weights of the old and new processes are equal, and consider rank-based distribution-free test statistics.

Suppose $H_0 : \gamma = 1$ is true, i.e. the two populations are equal. The observations $X_1, \ldots, X_m$ and $Y_1, \ldots, Y_n$ all come from the same distribution, so the null distribution of Wilcoxon's rank-sum statistic $W = \sum_{i=1}^{n} R_i$ of the ranks R_i of the "Y" observations in the pooled sample is the distribution-free Wilcoxon rank-sum distribution. But when the alternative is true, even if the

second population is less dispersed than the first ($\gamma < 1$), the rank sum W shows no tendency to become smaller or larger, so it cannot be used to test dispersion.

The alternative is to transform the observations not to ranks but to **scores** reflecting the difference in dispersion. If the "Y" observations are less dispersed than the "X" observations, they will cluster in the middle; so we can devise scores that are small for observations in the middle and large otherwise (or, conversely, large in the middle). The table below records, for the data of Example 4.10 (the new process is the second population), the ranks and several scores reflecting dispersion:

process	Old	Old	New	Old	New	New	Old	Old	New	New	Old
observation	103	105	109	110	112	113	115	117	118	119	120
rank	1	2	3	4	5	6	7	8	9	10	11
AB score	1	2	3	4	5	6	5	4	3	2	1
ST score	1	4	5	8	9	11	10	7	6	3	2
Mood score	25	16	9	4	1	0	1	4	9	16	25

The graph of ranks against the three scores (Figure 4.7 of the book) shows the difference clearly.

The **AB score** is the Ansari–Bradley score: score 1 to the smallest and largest ranks, 2 to the second smallest and second largest, and so on up to the middle rank. The **Ansari–Bradley statistic** T_{AB} is the sum of the Ansari–Bradley scores of the second sample; it tends to be large when the second population is less dispersed.

The **Siegel–Tukey (ST) score** is very similar to the Ansari–Bradley score, but devised so that without ties all ST scores are distinct. The Siegel–Tukey statistic T_{ST} is the sum of the Siegel–Tukey scores of the second sample. It was specially devised so that under the null its distribution coincides with the Wilcoxon rank-sum distribution.

The **Mood score** is the squared difference between each rank R_i and the middle rank $(N+1)/2$, $a_M(R_i) = (R_i - (N+1)/2)^2$; it was devised to be more sensitive to observations in the tails, making differences in scale easier to detect. The Mood statistic is also the sum of the Mood scores of the second sample.

Under $H_0 : \gamma = 1$ or $H_0 : f = g$, the null distributions of the Ansari–Bradley, Siegel–Tukey and Mood statistics are obtained as the permutation distributions of the scores. Under the lower-tailed alternative $H_1 : \gamma < 1$ (the second population is less dispersed), larger Ansari–Bradley and Siegel–Tukey statistics and smaller Mood statistics are more extreme, and the p-values come from the permutation distributions of the statistics. These p-values depend on the relative order of the observations but not on the actual values, so they are distribution-free. These distribution-free tests are the Ansari–Bradley, Siegel–Tukey and Mood tests.

Example 4.11

For the data of Example 4.10, the Ansari–Bradley statistic is $T_{AB} = 3+5+6+3+2 = 19$, the Siegel–Tukey statistic $T_{ST} = 5+9+11+6+3 = 34$, and the Mood statistic $T_M = 9+1+0+9+16 = 35$. Generate the exact permutation distribution of each statistic and test $H_0 : \gamma = 1$ against $H_1 : \gamma < 1$. First prepare the data:

```
> x = c(110, 115, 120, 105, 103, 117)
> y = c(109, 118, 113, 112, 119)
> curry = data.frame(label = c(rep("Old",6),rep("New",5)),
+                    x = c(x,y),rank = rank(c(x,y)))
> curry2 = cbind(curry %>% arrange(x),
+                AB = c(1:6, 5:1),
+                ST = c(1,4,5,8,9,11,10,7,6,3,2)) %>%
+   mutate(Mood = (rank-6)^2)
> head(curry2,3)
```

```
  label   x rank AB ST Mood
1   Old 103    1  1  1   25
2   Old 105    2  2  4   16
3   New 109    3  3  5    9
```

The three statistics are the sums of the scores of the observations with `label == "New"`:

```
> stats.obs = curry2 %>% filter(label == "New") %>% summarise(
    sum(AB),sum(ST),sum(Mood)) %>% unlist()
> stats.obs
 sum(AB)  sum(ST) sum(Mood)
      19       34        35
```

There are $\binom{11}{5} = 462$ shuffles, so we perform the exact permutation test. Generate the null distribution of each statistic and compute the p-values, taking care with the meaning of "extreme":

```
> N = 11; n = 5
> allperms = gtools::combinations(N,n) # perm. ids for "y" group
> stats.perm = apply(allperms,1,
+       function (perm.y.id){
+         curry2[perm.y.id,] %>% summarise(sum(AB),sum(ST),sum(
    Mood)) %>% unlist()
+       } )
> p.AB = mean(stats.perm[1,] >= stats.obs[1])
> p.ST = mean(stats.perm[2,] >= stats.obs[2])
> p.Mood = mean(stats.perm[3,] <= stats.obs[3])  # smaller is more extreme
> (p.vals = c(p.AB = p.AB, p.ST = p.ST, p.Mood = p.Mood))
     p.AB      p.ST    p.Mood
0.2294372 0.2683983 0.1861472
```

The three permutation p-values are all large, so $H_0 : \gamma = 1$ is not rejected. (Figure 4.8 of the book shows the null distributions of the statistics with the observed values.)

The Siegel–Tukey statistic was devised so that its null distribution is the Wilcoxon rank-sum distribution, a great advantage in the 1960s, when generating permutation distributions took days, but hardly an advantage now that computers are fast. Moreover the Siegel–Tukey scores are not exactly symmetric, so scoring from the right changes the value of the statistic slightly, and the same data can give different test results; it is now rarely used. Many other scores have been proposed besides these three, but the most basic and intuitive Ansari–Bradley test is the most frequently used test of scale.

In R, `ansari.test()`, `mood.test()` and `SiegelTukeyTest()` of the DescTools package perform approximate tests.

Example 4.12

For the data of Example 4.10, run the scale tests with `ansari.test()`, `mood.test()` and `SiegelTukeyTest()`, testing $H_0 : \gamma = 1$ against $H_1 : \gamma < 1$ as in Example 4.11. `ansari.test()` gives exact p-values for small samples.

```
> ansari.test(y,x,alternative = "less")

        Ansari-Bradley test

data:  y and x
AB = 19, p-value = 0.2294
alternative hypothesis: true ratio of scales is less than 1
```

`DescTools::SiegelTukeyTest()` computes the Siegel–Tukey statistic leaving out the median (here the value with ST score "11"). With such small samples this changes the p-value considerably.

```
> DescTools::SiegelTukeyTest(y,x,alternative = "less",exact =
    TRUE)

        Siegel-Tukey-test for equal variability

data:  y and x
ST = 23, p-value = 0.4571
alternative hypothesis: true ratio of scales is less than 1
```

Finally `mood.test()` computes the p-value from the standardized statistic with the large-sample approximation; for small samples the permutation p-value is more accurate.

```
> mood.test(y,x,alternative = "less")

        Mood two-sample test of scale

data:  y and x
Z = -0.98058, p-value = 0.1634
alternative hypothesis: less
```

4.4.2 When the two locations differ

If the two populations may differ in both dispersion and location,

$$X_i = \mu_1 + \sigma_1 e_i, \qquad Y_i = \mu_2 + \sigma_2 \epsilon_i,$$

how can we test $\gamma = \sigma_2/\sigma_1$ when $\mu_1 = \mu_2$ cannot be assumed?

If the populations can be assumed close to normal, the ratio of the two sample variances, using the estimators $S_1^2 = \sum_{i=1}^m (X_i - \bar{X})^2/(m-1)$ and $S_2^2 = \sum_{i=1}^n (Y_i - \bar{Y})^2/(n-1)$ of the population variances,

$$F = \frac{S_2^2}{S_1^2}, \tag{4.9}$$

follows the $F_{n-1,m-1}$ distribution (asymptotically) under H_0 ($\sigma_1^2 = \sigma_2^2$), which gives a test. (The book writes $F_{m-1,n-1}$; with S_2^2, from the n observations of Y, in the numerator, the degrees of freedom are $(n-1, m-1)$.)

When the populations are very different from normal, even if $H_0 : \gamma = 1$ is true, the two populations have different locations, so the exchangeability of the Ansari–Bradley permutation test (or of permutation tests with other scores) fails and the type I error cannot be controlled. The location-adjusted observations $\{X_i - \mu_1, Y_j - \mu_2 : i = 1, \ldots, m,\ j = 1, \ldots, n\}$ are exchangeable under H_0, but μ_1, μ_2 are unknown and cannot be used. Higgins (2004) proposed estimating the unknown locations and running an approximate permutation test with the location-adjusted observations $X_i - \hat{\mu}_1$, $Y_j - \hat{\mu}_2$.

First, some statistics usable when the populations are very different from normal but have the same shape apart from location and dispersion. The following three statistics tend to be large when the alternative $\gamma > 1$ is true:

1. the ratio F (4.9) of the two sample variances; with heavy tails or outliers its variability is large;

2. the ratio of median absolute deviations (MAD),

$$R = \frac{\text{median}\{|Y_i - \text{median}\{Y_i\}|\}}{\text{median}\{|X_i - \text{median}\{X_i\}|\}};$$

its variability is small even with heavy tails or outliers;

3. the Ansari–Bradley statistic of the median-adjusted observations: using that the median-adjusted observations $X_i' = X_i - \text{median}\{X_i\}$ and $Y_i' = Y_i - \text{median}\{Y_i\}$ have approximately the same location, compute the Ansari–Bradley statistic of $\{X_1', \ldots, X_m', Y_1', \ldots, Y_n'\}$. (As location estimate one may use the median of the Walsh averages instead of median$\{X_i\}$, and the Siegel–Tukey or Mood statistic instead of Ansari–Bradley.) Since T_{AB} (the sum of the second sample's scores) is large when the second sample is *less* dispersed, here it is small when $\gamma > 1$.

A permutation test can be run with each statistic. Under H_0 the locations differ, so $\{X_i, Y_j\}$ are not exchangeable, but the location-adjusted deviations $X_i' = X_i - \text{median}\{X_i\}$, $Y_j' = Y_j - \text{median}\{Y_j\}$ are (approximately) exchangeable, and shuffling them gives an approximate permutation test. (With the F statistic (4.9) use the mean $\bar{X}$ instead of the median median$\{X_i\}$ to form X_i'.)

The procedure of the approximate permutation test with the Ansari–Bradley statistic: first compute T_{AB} from the adjusted observations $\mathbb{X}' = \{X_1', \ldots, X_m', Y_1', \ldots, Y_n'\}$. Then, over all shuffles of the adjusted observations $\mathbb{X}'$ (or M random shuffles), compute the Ansari–Bradley statistic repeatedly, and use the distribution of $\{T_{AB}^{*i} : i = 1, \ldots, M\}$ as the null distribution. For $H_0 : \gamma = 1$ against $H_1 : \gamma > 1$, smaller T_{AB} is more extreme, so $p = M^{-1} \sum_{i=1}^{M} I(T_{AB}^{*i} \le T_{AB})$. (The book writes "$\ge$"; with T_{AB} the sum of the second sample's scores, $\gamma > 1$ makes it small.)

Example 4.13

For the acute leukemia data of Example 4.1, test whether the dispersions of ALL and AML differ. Vector `x` holds the ALL expression values and `y` the AML ones. First, the F-test with the ratio F (4.9) of the sample variances gives a fairly large p-value, not enough evidence that the variances differ. The samples are large enough to trust the F-test.

```
> var.test(x,y)$p.value
[1] 0.1986019
```

Next, median-adjusted statistics. The two locations cannot be assumed equal, so instead of the Ansari–Bradley test directly we must use the median-adjusted observations:

```
> ansari.test(x - median(x), y - median(y))

        Ansari-Bradley test

data:  x - median(x) and y - median(y)
AB = 933, p-value = 0.1347
alternative hypothesis: true ratio of scales is not equal to 1
```

Without median adjustment the differences in scale and location are confounded and cannot be distinguished. A wrong example:

```
> ansari.test(x,y)

        Ansari-Bradley test

data:  x and y
AB = 958.5, p-value = 0.03518
alternative hypothesis: true ratio of scales is not equal to 1
```

The Siegel–Tukey function `SiegelTukeyTest()` has an argument for median adjustment; the two calls below give the same result.

```
DescTools::SiegelTukeyTest(x,y,adjust.median = TRUE)
DescTools::SiegelTukeyTest(x - median(x), y - median(y))
```

4.5 The Kolmogorov–Smirnov test

So far we have looked at tests of location and dispersion of two populations. This section examines the **Kolmogorov–Smirnov** (KS) test, which tests a broad alternative of difference between two populations without fixing one parameter (location or dispersion).

With distribution functions F, G of the two populations and observations $X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} F$, $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} G$, we test

$$H_0 : F = G \quad \text{vs.} \quad H_1 : F \neq G.$$

The alternative is not written in terms of a particular parameter, so under it G and F may differ in shape even with equal means and variances (e.g. G more right-skewed than F). Then permutation tests using the Wilcoxon rank-sum or Ansari–Bradley statistic, sensitive to the shift Δ or scale γ, have very low power and may fail to reject H_0 even when the distributions are very different.

The Kolmogorov–Smirnov statistic directly compares the **empirical distribution functions** obtained from each sample. First, the empirical distribution function $\hat{F}_X$ of a sample $X_1, \ldots, X_m$. The (cumulative) distribution function of a random variable X (or a population) gives for each $x \in (-\infty, \infty)$ the probability $F(x) = P(X \leq x)$ that $X \leq x$. The true F is unknown, so a good estimator of $F(x)$ is the proportion of observations at most x,

$$\hat{F}_X(x) = \hat{P}(X \leq x) = \frac{1}{m} \sum_{i=1}^{m} I(X_i \leq x),$$

the empirical distribution function. E.g. Figure 4.9 of the book compares the empirical distribution function of $m = 40$ observations randomly drawn from the standard normal with the standard normal distribution function.

At each x, $\hat{F}_X(x)$ lies around the true $F_X(x)$. Indeed $E(\hat{F}_X(x)) = F_X(x)$ and $\text{Var}(\hat{F}_X(x)) = F_X(x)(1 - F_X(x))/n$ (here $n = m$, the sample size), so for large samples the empirical distribution function is very close to the true one.

To compare the distributions of two samples, compare their empirical distribution functions. Figure 4.10 of the book shows those of the ALL and AML samples of the acute leukemia patients of Example 4.1: AML values tend to be larger than ALL values, and the larger the difference of the two empirical distribution functions (the vertical distance between the lines), the more we believe the distributions differ.

The **Kolmogorov–Smirnov statistic** (KS statistic) K is the maximal difference of the two empirical distribution functions $\hat{F}_X(x)$ and $\hat{F}_Y(x)$, scaled using the sample sizes:

$$K := \sqrt{\frac{mn}{m+n}} \sup_x \{|\hat{F}_X(x) - \hat{F}_Y(x)|\}.$$

Under $H_0 : F = G$ both $\hat{F}_X(x)$ and $\hat{F}_Y(x)$ approach $F(x)$, so the statistic is multiplied by $\sqrt{\frac{mn}{m+n}} = \left(\frac{1}{m} + \frac{1}{n}\right)^{-1/2}$ to keep it from becoming too small. The two empirical distribution functions are step functions with at most $m+1$ and $n+1$ distinct values, so to compute K we need not examine the difference at every x, only at each observation X_i, Y_j. When the populations differ (the alternative

is true), whatever the form of the difference, K tends to be large, so the KS statistic tests any form of alternative. But because the KS test is sensitive to every possible alternative, it is less powerful than tests sensitive only to specific alternatives (e.g. the Wilcoxon rank-sum test, sensitive only to location differences).

The KS test rejects H_0 when K is large. The null distribution of K is known to depend only on m, n, whatever the null distribution $F = G$: the KS test is distribution-free. The exact null distribution of K is complicated to describe, but if the null distribution $F = G$ is continuous, its large-sample distribution is the Kolmogorov distribution.[6] R's `ks.test()` uses the exact null distribution of K for small samples without ties, and otherwise the Kolmogorov distribution, the large-sample distribution of K. Alternatively a (random) permutation test with the KS statistic K can be run.

Example 4.14

For the two empirical distribution functions of the acute leukemia patients of Example 4.1, the maximal difference is $D = \sup_x\{|\hat{F}_X(x) - \hat{F}_Y(x)|\} = \max_{i,j}\{|\hat{F}_X(z) - \hat{F}_Y(z)| : z = X_i \text{ or } Y_j\} = 0.41617$, so the observed KS statistic is $K_{\text{obs}} = \sqrt{\frac{mn}{m+n}}D = \sqrt{\frac{47\times25}{47+25}} \times 0.41617 \approx 1.68$. Compared with the Kolmogorov distribution, the large-sample distribution of K (Figure 4.11 of the book), $p = P(K \geq K_{\text{obs}}) \approx 0.007$: strong evidence that the gene expression of the two types of acute leukemia patients differs. R's `ks.test()` approximates the p-value simply; vector `x` holds the ALL values and `y` the AML values.

```
> ks.test(x,y)

        Two-sample Kolmogorov-Smirnov test

data:  x and y
D = 0.41617, p-value = 0.007014
alternative hypothesis: two-sided
```

4.6 Two-sample tests: which test to use?

Which test has the highest power while controlling the type I error when comparing the locations of two populations? In nonparametric statistics, with its very broad range of populations, the answer is not simple. The Wilcoxon–Mann–Whitney test, or permutation tests, require the two distributions to be equal under the null and can be used only when this assumption holds. In this section, for tests of the locations of two populations, we treat separately the case where the distributions are equal under the null and the case where under the null the locations are equal but the dispersions differ.

4.6.1 Comparison in the location-shift model

Assume observations $X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} F$, $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} G$ and a shift Δ with

$$X_i \overset{d}{=} Y_j - \Delta.$$

That is, the mean (or median) of G exceeds that of F by Δ, a location-shift model in which the difference of the two populations is described only by Δ. When $H_0 : \Delta = 0$ is true, the two distributions are equal.

[6] If the population $F = G$ under the null is continuous, the asymptotic distribution of the KS statistic is $\lim_{m,n\to\infty} P(K \leq x) = 1 - 2\sum_{j=1}^{\infty}(-1)^{j-1}\exp(-2j^2x^2)$, the Kolmogorov distribution.

We compare four tests: (1) the two-sample t-test, (2) the Wilcoxon–Mann–Whitney rank-sum test, (3) the permutation test with the two-sample t-statistic (t-permutation test) and (4) the permutation test with the difference of medians (m-permutation test).

Type I error. All four tests control the type I error approximately below α. The nonparametric rank-sum test and the permutation tests are theoretically exact, so they control the type I error for any population. The two-sample t-test controls it exactly only for normal populations; for populations with thinner tails than normal (e.g. uniform) its type I error is not much larger than α, and for large m, n it controls it approximately. (The same held for one-sample tests of location.) For heavy-tailed or asymmetric populations (e.g. Laplace or exponential) the two-sample t-test is rather conservative (type I error $< \alpha$).

Power comparison. Before comparing, recall that for large samples the two-sample t-test and the permutation test with the two-sample t-statistic are similar. The table below compares which test is most powerful in which situation (Keller-Mcnulty and Higgins, 1987; Higgins, 2004).

	small samples	large samples
thin tails (uniform)	t-perm. $>$ t-test $>$ Wilcoxon	t-test $\approx$ t-perm.
normal	t-test	t-test $\approx$ t-perm.
heavy tails (Laplace)	Wilcoxon $>$ t-test $\approx$ t-perm.[7]	Wilcoxon
very heavy tails (Cauchy)	m-perm. $>$ Wilcoxon $>$ t-test	m-perm.

Small samples are $m + n = 12$ and large samples $m + n = 36, 128$ (same conclusions in both cases). For very large samples the comparison can be summarized by the asymptotic relative efficiency. The ARE of the Wilcoxon–Mann–Whitney test relative to the two-sample t-test:

	ARE(Wilcox, t)
uniform	1
normal	$3/\pi \approx 0.955$
Laplace	1.5
exponential	3
Cauchy	∞

(The book prints 0.995 for the normal; the value is $3/\pi \approx 0.955$, as for the signed-rank test in Section 2.3.2. In the shift model the ARE is $12\sigma^2(\int f^2)^2$, giving these entries.) Except for the normal, the Wilcoxon rank-sum test has higher ARE, and even when normality holds its efficiency is not much lower than that of the t-test.

4.6.2 When the two distributions differ under the null hypothesis

Suppose the means and variances of the two populations are (μ_1, σ_1^2), (μ_2, σ_2^2) and the variances cannot be assumed equal:

$$\sigma_1^2 \neq \sigma_2^2.$$

Then even when the null hypothesis $H_0 : \mu_1 = \mu_2$ about location is true, the two distributions differ, so neither the Wilcoxon–Mann–Whitney test nor any permutation test is guaranteed to control the type I error.

[7] "Wilcoxon $>$ t-test $\approx$ t-perm." means the Wilcoxon test is more powerful than the t-test, whose power is similar to that of the t-permutation test. The m-permutation test, not listed, is less powerful than the other tests.

If the populations follow normality approximately, or the samples are large enough, Welch's two-sample t-test can be used. If the samples are not large enough or normality is badly violated, first try a transformation of the variables (e.g. $x \mapsto \log(x)$) so that the transformed samples approximately follow normality. As a rank-based approximate distribution-free method, a standardized Wilcoxon rank-sum statistic as in Chung and Romano (2016) can be used; comparative studies of the power of such methods are insufficient and are not discussed here.